

37 ☆

```
char c = -1;
```

38

```
signed char d = 0;
```

39

```
printf("%d ", ~d);
```

```
printf("%d ", --d);
```

40

```
printf("%d ", d || c);
```

```
return 0;
```

```
}
```

☐ -1 -1 0

☒ -1 -1 1

☐ 1 1 1

☐ 1 -1 0

int main()

{

int a, b;

a = 33;

b = 10;

(a % b == 0) ? printf("%d", a += 10) : printf("%d", (b -= 10));

return 0;

}

☐ 23

☐ 20

☐ 43

☐ 0

Mark(s): 1

A binary tree having  $n$  nodes and depth  $d$  will be complete binary tree if:

- I) any node  $n_d$  at level less than  $d-1$  has two children.
- II) it contains  $2^{d+1}-1$  nodes.
- III) for any node  $n_d$  in the tree with a right descendent at level  $d$ ,  $n_d$  must have a left child.

☐ All (I), (II) and (III)

☐ Only (II)

☐ Only (III)

☐ Only (I)

Which of the below given switching technique(s) is/are more reliable?

- I) Circuit switching
- II) Message switching
- III) Packet switching

☒ Only (III)

☐ Only (I) and (II)

☐ Only (I) and (III)

☐ All (I), (II) and (III)

QUESTION: 25    GROUP: Technical    SECTION: Operating System

Mark(s): 1

Which of the given below statement(s) is/are not true regarding page table length?

- I) Page table is of variable length.
- II) Page table length is depends upon the size of the process.

☐ Both (I) and (II)

☐ Only (II)

☐ Only (I)

☐ Neither (I) nor (II)

14 QUESTION: 18 GROUP: Technical SECTION: Operating System  
Mark(s): 1

15 What will be the displacement into the page for the generated virtual address 90904141 if the virtual memory provided to the user is  $2^{32}$   
16 physical memory available to the user is  $2^{18}$  bytes?  
7

☐ 0001 0100 0001

☐ 1001 0000 0100

☐ 1001 0000 1001

☐ 0100 0001 0100

☆

`printf("%d", x);`

`return 0;`

`}`

☐ 6

☐ 7

☐ 5

☐ 4

QUESTION: 36    GROUP: C    SECTION: C

Mark(s): 1

A pointer to a function that takes two floats and returns a pointer to a pointer to a char would be declared as:

☐ `char ** (*p)(float, float);`

☒ `float *(*p) (char, char);`

☐ `char *(*p)(float, float);`

☐ `char **p(float, float);`



return(w);

int sum(int p, int q)

{

return p & q << p - 1;

}



Compile time error at //Statement 1.



Compile time error at //Statement 2.



Compile time error as function's declaration are missing.



No error in the program.

What will be the output of the program given below?

```
#include <stdio.h>

int main(void)
{
    char a = 6;

    char b = -3;

    printf("%d ", a % b);

    printf("%d ", a % -b);

    printf("%d ", a / b);

    printf("%d ", a / -b);

    return 0;
}
```

QUESTION: 13    GROUP: Technical    SECTION: Operating System

Mark(s): 1

Which one of the given options is the major application of the fair share scheduling?

☐ Guaranteed Scheduling

☐ Gang Scheduling

☐ Real Time Scheduling

☐ Thread Scheduling



Mark(s): 1

Which of the given below statement(s) is/are not true regarding page table length?

I) Page table is of variable length.

II) Page table length is depends upon the size of the process.

☐ Both (I) and (II)

☐ Only (II)

☐ Only (I)

☐ Neither (I) nor (II)

QUESTION: 21 GROUP: Technical SECTION: Data Structures

$k(s)$ : 1

will be the maximum possible height of a 4-arr tree whose root has four children and total 30 nodes?

(Assume height of a single node is 0.)



QUESTION: 21    GROUP: Technical    SECTION: Data Structures

Mark(s): 1

What will be the maximum possible height of a 4-arr tree whose root has four children and total 30 nodes?

[Note: Assume height of a single node is 0.]

☐ 5

☐ 26

☐ 24

☐ 4



Which of the given below statement(s) is/are false in the context of anonymous class?

- I) An anonymous inner class cannot have an explicit constructor declaration.
- II) An anonymous inner class cannot have static members.
- III) An anonymous inner class cannot be abstract.
- IV) The class instance creation expression for an anonymous class must never include arguments.

☐ Only (I) and (IV)

☐ Only (I)

☐ Only (II)

☐ Only (II) and (IV)



Mark(s): 1

Which of the below given statement(s) is/are correct?

- I) Use of functions is allowed inside structures in C++.
- II) Use of array is allowed inside structures in C++.

☐ Neither (I) nor (II)

☒ Only (II)

☐ Both (I) and (II)

☐ Only (I)



QUESTION: 40 GROUP: C SECTION: C

Mark(s): 1

Referring to the sample below, which of the statements given in the options is correct?

```
if (x ? y: z) doSomething();
```

☐ y is evaluated only when  $x == 0$

☐ y is evaluated only when  $x \geq 1$

☐ y is evaluated only when  $x \neq 0$

☐ y is evaluated only when  $x == 1$

```
float t;  
short int y;  
};  
};  
p;  
printf("%d", sizeof(u) + sizeof(p.s) + sizeof(p));  
return 0;  
}
```

☐ 8

☐ Compile time error

☐ 10

☐ 20

Mark(s): 10

Write a program to determine the number of times  $T$  a given number  $A$  consisting of  $N$  digits or the anagrams of  $A$  occur in another number  $B$  consisting of  $M$  digits in a single iteration.

An anagram of a number is formed by using exactly the same digits of the original number but with a different arrangement. For example, anagrams of the number 123 are: 321, 312, 213, 231 and 132.

Input from STDIN and the output from STDOUT. You should not write arbitrary strings while reading the input and while printing as these are the standard output.

**Constraints:**

- I)  $A$  and  $B$  are positive integers.
- II)  $0 < N, M < 100$

**Input Format:**

The first line of the input contains  $A$  in the form of a string.

The second line of the input contains  $B$  in the form of a string.

**Output:**

The output consists of  $T$ .

**Sample Input 1:**

23

23212321

**Sample Output 1:**

8 ☆

```
};  
union test *t;  
t = (union test *)malloc(sizeof(union test));  
t->f = 10.10f;  
printf("%f", t->f);  
return 0;  
}
```

☐ 10

☐ Garbage value

☐ Compile time error

☐ 10.100000

19 3. If  $\text{Item} = A[\text{Mid}]$  then  
20 4. Return Mid  
21 5. If  $\text{Item} > A[\text{Mid}]$  then  
22 6. Return BSEARCH (A, Mid + 1, High)  
23 7. Else  
24 8. Return BSEARCH (A, Low, Mid - 1)  
25 End If Step 1  
26 9. Return -1

☐ High

☐  $(\text{High} + \text{Low}) / 2$

☐  $(\text{High} - \text{Low}) / 2$

☐ High - Low



QUESTION: **12**      GROUP: **Technical**      SECTION

Mark(s): 1

How many bits are there in the physical address, if

☐ 15

☐ 9

☐ 12

☐ 14

Mark(s): 1

Consider a segmentation system with the segment table given below:

| Base address | Length in bytes |
|--------------|-----------------|
| 620          | 105             |
| 165          | 160             |
| 231          | 220             |

What will be the physical addresses for the given logical addresses? Will a segmentation fault occur for any of the segments?

I) 0, 40

II) 1, 60

☐ 660, 291

☐ 40, 60

☐ Segmentation faults occur for both addresses.

☐ 145, 220